

Caleb Pickens

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EDUCATION

The University of Texas at Austin, Austin, TX

Expected May 2028

Bachelor of Science, Computer Science, GPA 3.62

Relevant Coursework: Operating Systems, Computer Organization & Architecture, Data Structures & Algorithms

SKILLS

Languages: Python, Java, JavaScript, HTML, CSS (Familiar with C and C++)

Frameworks & Libraries: Node.js (Familiar with React)

Developer Tools: Git, Linux, REST APIs

PROJECTS

VACUUM REGION IDENTIFICATION ALGORITHM | Freshman Research Initiative

- Developed a Python and Fortran algorithm to identify and group vacuum regions in 3D electron density data.
- Collaborated with a partner to test and validate the algorithm for characterizing material structural features.

HUFFMAN FILE COMPRESSION | Data Structures & Algorithms

- Engineered a file compression and decompression program in Java using the Huffman coding algorithm.
- Implemented a custom fair priority queue to build an optimized Huffman tree based on character frequency.
- Developed bit-level I/O operations to write compressed data and file headers to recreate the Huffman tree.

PROCESSOR PIPELINE SIMULATOR | Computer Organization & Architecture

- Developed a five-stage MIPS processor pipeline simulator in C to parse and execute a subset of the MIPS instruction set.
- Implemented forwarding and stalling logic to resolve data and control hazards, ensuring correct program execution.

EXPERIENCE

UNIVERSITY OF TEXAS, Austin, TX

August 2025 - Present

Undergraduate Course Assistant

- Mentor 300+ students in Python fundamentals, providing personalized feedback and debugging assistance.
- Analyze student code to diagnose logical errors and communicate complex programming concepts.
- Collaborate with faculty to ensure fair and consistent grading for a large-scale course.

IDEA LAB KIDS, Austin, TX

January 2025 - May 2025

STEM Instructor

- Developed and delivered a hands-on curriculum in Python and Java to introduce object-oriented programming and core computer science principles to students aged 8-12.
- Simplified complex software development concepts into intuitive, engaging lessons, demonstrating a passion for making technology accessible and easy to understand.

SURE LEADER LLC, Remote Work

September 2020 - March 2021

Freelance Website Developer

- Engineered and launched user-focused, responsive websites for over 20 clients, prioritizing intuitive navigation and a clean user experience from conception to deployment.
- Managed the full project lifecycle, from initial client consultation and requirements gathering to deployment.
- Fostered strong client relationships through clear communication and timely project delivery, resulting in a high rate of repeat business and referrals.

COMMUNITY INVOLVEMENT

NEW LIFE CHILDREN'S CENTER, Canyon Lake, TX

Volunteer

June 2018 - Present

- Provide mentorship and a supportive, positive environment for teenagers in residential care.
- Develop digital content, including videos and web pages, to enhance community outreach and support fundraising initiatives.